

# ANX-PR/CL/001-01

## LEARNING GUIDE

### SUBJECT

**103000739 - Adaptive Systems**

### DEGREE PROGRAMME

**10AM - Master Universitario En Ingenieria Del Software**

### ACADEMIC YEAR & SEMESTER

**2025/26 - Semester 1**

## Index

---

### Learning guide

|                                                         |    |
|---------------------------------------------------------|----|
| 1. Description.....                                     | 1  |
| 2. Faculty.....                                         | 1  |
| 3. Prior knowledge recommended to take the subject..... | 2  |
| 4. Skills and learning outcomes .....                   | 2  |
| 5. Brief description of the subject and syllabus.....   | 3  |
| 6. Schedule.....                                        | 5  |
| 7. Activities and assessment criteria.....              | 8  |
| 8. Teaching resources.....                              | 14 |
| 9. Other information.....                               | 15 |

## 1. Description

### 1.1. Subject details

|                                       |                                                        |
|---------------------------------------|--------------------------------------------------------|
| <b>Name of the subject</b>            | 103000739 - Adaptive Systems                           |
| <b>No of credits</b>                  | 4 ECTS                                                 |
| <b>Type</b>                           | Optional/elective                                      |
| <b>Academic year of the programme</b> | First year                                             |
| <b>Semester of tuition</b>            | Semester 1                                             |
| <b>Tuition period</b>                 | September-January                                      |
| <b>Tuition languages</b>              | English                                                |
| <b>Degree programme</b>               | 10AM - Master Universitario en Ingeniería del Software |
| <b>Centre</b>                         | 10 - E.T.S. De Ingenieros Informáticos                 |
| <b>Academic year</b>                  | 2025-26                                                |

## 2. Faculty

### 2.1. Faculty members with subject teaching role

| <b>Name and surname</b>                          | <b>Office/Room</b> | <b>Email</b>              | <b>Tutoring hours *</b>                                                                                                                            |
|--------------------------------------------------|--------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Angelica De Antonio Jimenez                      | 5108               | angelica.deantonio@upm.es | M - 10:30 - 14:00<br>Th - 09:30 - 12:00                                                                                                            |
| Jaime Ramirez Rodriguez<br>(Subject coordinator) | 5109               | jaime.ramirez@upm.es      | Sin horario.<br>The tutoring table<br>can be found at: <a href="https://dlsiis.fi.upm.es/tutorias-2526">https://dlsiis.fi.upm.es/tutorias-2526</a> |

\* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

### 3. Prior knowledge recommended to take the subject

---

#### 3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

#### 3.2. Other recommended learning outcomes

- Computer programming

### 4. Skills and learning outcomes \*

---

#### 4.1. Skills to be learned

CE13 - Tener una visión de los distintos aspectos específicos y emergentes de la ingeniería del software, y profundizar en algunos de ellos

CE14 - Comprender lo que pueden y no pueden conseguir las prácticas actuales de ingeniería del software, y sus limitaciones y su posible futura evolución.

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

CG3 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades (RD)

CG8 - Planteamiento y resolución de problemas también en áreas nuevas y emergentes de su disciplina

CG9 - Aplicación de los métodos de resolución de problemas más recientes o innovadores y que puedan implicar el uso de otras disciplinas

## 4.2. Learning outcomes

RA88 - Knowledge of Methods for student modelling and individualized and adapted interaction with learning systems

RA89 - Model the user and to design adaptive user interfaces based on the user

\* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

## 5. Brief description of the subject and syllabus

---

### 5.1. Brief description of the subject

Interaction design methods focused on providing the same content to all users offer limited possibilities in addressing the specific needs and requirements of different types of users. Therefore, an essential feature of certain interactive applications should be their ability to provide some form of automatic adaptation and customization. The main objective of this course is to achieve an understanding of the models, techniques and architectures necessary to make a computer application dynamically adapt to the specific needs and requirements of different types of users at all times.

Adaptive systems maintain a model of the interests, preferences and / or knowledge of each individual user, and use this model to adapt the behavior of the systems to the needs of that user.

This course will cover the main components of the user model in the context of adaptive systems. And within the adaptive systems, the personalized search systems on the Web will be addressed first and it will be explained how these systems are supported by the user model.

Recommendation systems have become essential tools in many areas of application, because they help alleviate information overload as they select the most appropriate content for each user based on their preferences and / or interests. In this sense, these types of systems help users in decision-making by providing personalized services and help information providers and companies to serve customers more effectively.

In this course, the general characteristics of the recommendation systems will be explained and a classification of these systems will be presented according to the type of techniques they use to generate the recommendations. According to this classification, we will distinguish between content-based, collaborative filtering, and hybrid recommenders. Following this, an introduction will be given to the main techniques on which each of these groups of recommenders are based, highlighting the strengths and weaknesses of each group.

E-learning is a traditional domain for the application of personalization and adaptation technologies. One of the main objectives of these applications is to improve the effectiveness and efficiency of learning experiences. The last topic of the course will be about adaptive e-learning systems paying special attention to intelligent tutoring systems.

## 5.2. Syllabus

1. User Modeling for Adaptive Systems and Adaptive Web
2. Recommender Systems
  - 2.1. Content based Recommender Systems
  - 2.2. Collaborative Filter based Recommender Systems
  - 2.3. Context aware Recommender Systems
  - 2.4. Deep learning applied to Recommender Systems
3. Technology-enhanced adaptive learning

## 6. Schedule

### 6.1. Subject schedule\*

| Week | Type 1 activities                                                                                 | Type 2 activities | Distant / On-line | Assessment activities                                                                                                                                                                                                                                      |
|------|---------------------------------------------------------------------------------------------------|-------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Presentation of the subject</b><br>Duration: 02:00<br>Lecture                                  |                   |                   | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                        |
| 2    | <b>User modeling for Adaptive Systems and Adaptive Web</b><br>Duration: 02:00<br>Lecture          |                   |                   | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                        |
| 3    | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |                   |                   | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                        |
| 4    | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |                   |                   | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                        |
| 5    | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |                   |                   | <b>Exam 1</b><br>Online test<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 01:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00              |
| 6    | <b>Workshop associated with practical exercise 1</b><br>Duration: 02:00<br>Laboratory assignments |                   |                   | <b>Practical exercise 1</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 07:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |

|    |                                                                                                   |  |  |                                                                                                                                                                                                                                                            |
|----|---------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |  |  | <b>Practical exercise 1</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 07:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |
| 8  | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |  |  | <b>Practical exercise 1</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 20:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |
| 9  | <b>Workshop associated with practical exercise 2</b><br>Duration: 02:00<br>Laboratory assignments |  |  | <b>Practical exercise 2</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 20:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |
| 10 | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |  |  | <b>Practical exercise 2</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 10:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |
| 11 | <b>Recommender Systems</b><br>Duration: 02:00<br>Lecture                                          |  |  | <b>Practical exercise 2</b><br>Group work<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 10:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00 |



|    |                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | <b>Technology-enhanced adaptive learning</b><br>Duration: 02:00<br>Lecture                                                                                                                                       |  |  | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00<br><br><b>Exam 2</b><br>Online test<br>Progressive assessment and Global Examination<br>Not Presential<br>Duration: 01:00                                                                                                                                                          |
| 13 | <b>Technology-enhanced adaptive learning</b><br>Duration: 02:00<br>Lecture                                                                                                                                       |  |  | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                                                                                                                                                                    |
| 14 | <b>Technology-enhanced adaptive learning</b><br>Duration: 01:00<br>Problem-solving class<br><br><b>Technology-enhanced adaptive learning</b><br>Duration: 01:00<br>Lecture                                       |  |  | <b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00                                                                                                                                                                                                                                                                                    |
| 15 | <b>Technology-enhanced adaptive learning (student presentations)</b><br>Duration: 01:30<br>Additional activities<br><br><b>Technology-enhanced adaptive learning</b><br>Duration: 00:30<br>Problem-solving class |  |  | <b>Design of Adaptive Tutoring System</b><br>Group work<br>Progressive assessment<br>Not Presential<br>Duration: 00:00<br><br><b>Classroom Participation grade</b><br>Other assessment<br>Progressive assessment<br>Presential<br>Duration: 00:00<br><br><b>Technology-enhanced adaptive learning (student presentations)</b><br>Group work<br>Progressive assessment<br>Presential<br>Duration: 01:30 |
| 16 |                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                        |
| 17 |                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                        |

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

## 7. Activities and assessment criteria

### 7.1. Assessment activities

#### 7.1.1. Assessment

| Week | Description                   | Modality         | Type          | Duration | Weight | Minimum grade | Evaluated skills                          |
|------|-------------------------------|------------------|---------------|----------|--------|---------------|-------------------------------------------|
| 1    | Classroom Participation grade | Other assessment | Face-to-face  | 00:00    | .33%   | 0 / 10        | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13 |
| 2    | Classroom Participation grade | Other assessment | Face-to-face  | 00:00    | .33%   | 0 / 10        | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13 |
| 3    | Classroom Participation grade | Other assessment | Face-to-face  | 00:00    | .33%   | 0 / 10        | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13 |
| 4    | Classroom Participation grade | Other assessment | Face-to-face  | 00:00    | .33%   | 0 / 10        | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13 |
| 5    | Exam 1                        | Online test      | No Presential | 01:00    | 5%     | 0 / 10        | CG13<br>CE14<br>CE13                      |
| 5    | Classroom Participation grade | Other assessment | Face-to-face  | 00:00    | .33%   | 0 / 10        | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13 |

|    |                               |                  |               |       |      |        |                                            |
|----|-------------------------------|------------------|---------------|-------|------|--------|--------------------------------------------|
| 6  | Practical exercise 1          | Group work       | No Presential | 07:00 | 5%   | 5 / 10 | CG13<br>CE14<br>CG3<br>CE13                |
| 6  | Classroom Participation grade | Other assessment | Face-to-face  | 00:00 | .33% | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 7  | Practical exercise 1          | Group work       | No Presential | 07:00 | 5%   | 5 / 10 | CG13<br>CE14<br>CG3<br>CE13                |
| 7  | Classroom Participation grade | Other assessment | Face-to-face  | 00:00 | .33% | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 8  | Practical exercise 1          | Group work       | No Presential | 20:00 | 15%  | 5 / 10 | CG13<br>CE14<br>CG3<br>CE13                |
| 8  | Classroom Participation grade | Other assessment | Face-to-face  | 00:00 | .33% | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 9  | Practical exercise 2          | Group work       | No Presential | 20:00 | 20%  | 5 / 10 | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |
| 9  | Classroom Participation grade | Other assessment | Face-to-face  | 00:00 | .33% | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 10 | Practical exercise 2          | Group work       | No Presential | 10:00 | 10%  | 5 / 10 | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |

|    |                                    |                  |               |       |       |        |                                            |
|----|------------------------------------|------------------|---------------|-------|-------|--------|--------------------------------------------|
| 10 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .33%  | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 11 | Practical exercise 2               | Group work       | No Presential | 10:00 | 5%    | 5 / 10 | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |
| 11 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .33%  | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 12 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .33%  | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 12 | Exam 2                             | Online test      | No Presential | 01:00 | 5%    | 0 / 10 | CG9<br>CG13<br>CE14<br>CE13                |
| 13 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .33%  | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 14 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .33%  | 0 / 10 | CG9<br>CG13<br>CG3<br>CG8<br>CG18<br>CE13  |
| 15 | Design of Adaptive Tutoring System | Group work       | No Presential | 00:00 | 12.5% | 5 / 10 | CE14<br>CG18<br>CE13<br>CG9<br>CG13        |
| 15 | Classroom Participation grade      | Other assessment | Face-to-face  | 00:00 | .38%  | 0 / 10 |                                            |

|    |                                                               |            |              |       |       |        |                             |
|----|---------------------------------------------------------------|------------|--------------|-------|-------|--------|-----------------------------|
| 15 | Technology-enhanced adaptive learning (student presentations) | Group work | Face-to-face | 01:30 | 12.5% | 5 / 10 | CG13<br>CE14<br>CG3<br>CE13 |
|----|---------------------------------------------------------------|------------|--------------|-------|-------|--------|-----------------------------|

### 7.1.2. Global examination

| Week | Description          | Modality    | Type          | Duration | Weight | Minimum grade | Evaluated skills                           |
|------|----------------------|-------------|---------------|----------|--------|---------------|--------------------------------------------|
| 5    | Exam 1               | Online test | No Presential | 01:00    | 5%     | 0 / 10        | CG13<br>CE14<br>CE13                       |
| 6    | Practical exercise 1 | Group work  | No Presential | 07:00    | 5%     | 5 / 10        | CG13<br>CE14<br>CG3<br>CE13                |
| 7    | Practical exercise 1 | Group work  | No Presential | 07:00    | 5%     | 5 / 10        | CG13<br>CE14<br>CG3<br>CE13                |
| 8    | Practical exercise 1 | Group work  | No Presential | 20:00    | 15%    | 5 / 10        | CG13<br>CE14<br>CG3<br>CE13                |
| 9    | Practical exercise 2 | Group work  | No Presential | 20:00    | 20%    | 5 / 10        | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |
| 10   | Practical exercise 2 | Group work  | No Presential | 10:00    | 10%    | 5 / 10        | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |
| 11   | Practical exercise 2 | Group work  | No Presential | 10:00    | 5%     | 5 / 10        | CG9<br>CG13<br>CE14<br>CG8<br>CG18<br>CE13 |
| 12   | Exam 2               | Online test | No Presential | 01:00    | 5%     | 0 / 10        | CG9<br>CG13<br>CE14<br>CE13                |

### 7.1.3. Referred (re-sit) examination

| Description                        | Modality        | Type         | Duration | Weight | Minimum grade | Evaluated skills                                  |
|------------------------------------|-----------------|--------------|----------|--------|---------------|---------------------------------------------------|
| Practical exercise 1               | Individual work | Face-to-face | 20:00    | 30%    | 5 / 10        | CG9<br>CG13<br>CE14<br>CG3<br>CG8<br>CG18<br>CE13 |
| Design of Adaptive Tutoring System | Individual work | Face-to-face | 02:00    | 12.5%  | 5 / 10        | CG13<br>CE14<br>CE13                              |
| Practical exercise 2               | Individual work | Face-to-face | 40:00    | 45%    | 5 / 10        | CG9<br>CG13<br>CE14<br>CG3<br>CG8<br>CG18<br>CE13 |
| Research topic analysis            | Individual work | Face-to-face | 00:00    | 12.5%  | 5 / 10        | CE14<br>CG3<br>CE13<br>CG9<br>CG13                |

## 7.2. Assessment criteria

### 1. Regular Period

#### 1.1. Distributed evaluation

The final grade (FG) will be calculated from the grades of the Practical Exercises (PEG1, PE2G), the Exam grade (EG), the Classroom Participation grade (CPG), the grade of the Design of Adaptive Tutoring System (DTG), and the Research Topic Analysis grade (RTG) by means of the following formula:

$$FG=0.25*PEG1+ 0.35*PEG2 + 0.10*EG + 0.125*RTG + 0.125*DTG + 0.05CPG$$

Where all the grades take values between 0 and 10.

The Exam grade is the average of the grades obtained in the two exams.

The classroom participation grade stands for the ratio of attendance multiplied by 10. Additionally, **students will have to attend some prefixed classes associated with evaluation activities, which will be reported at least 15 days prior to the day of the evaluation activity.**

The grades obtained in the passed parts of the subject will be recorded for the global evaluation and the Extraordinary period if needed.

## 1.2. Global evaluation

If the student fails to pass the exam grade of the distributed evaluation, he/she will have a second chance in the global evaluation by doing a remedial exam, which will determine the exam grade (EG) of the regular period.

If the student fails to pass some of the practical exercises, he/she will have a second chance in the global evaluation to submit the failed practical exercises.

However, given the proximity of the deadline of the research topic analysis and the Design of the Adaptive Tutoring to the global evaluation, it does not make sense that students can submit these assignments in the global evaluation.

## 2. Extraordinary Period

When failed, in the extraordinary period the final grade will be obtained from the grades of the Practical Exercises (30% + 45%), the grade of the Design of Adaptive Tutoring System (12.5%), and the Research Topic Analysis grade (12.5%).

## 8. Teaching resources

### 8.1. Teaching resources for the subject

| Name      | Type         | Notes                                                                                                                                                                                                                        |
|-----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article   | Bibliography | Brusilovsky, Peter, and Eva Millán. 2007. "User Models for Adaptive Hypermedia and Adaptive Educational Systems". The Adaptive Web, 3-53.<br>doi:10.1007/978-3-540-72079-9_1.                                                |
| book      | Bibliography | Ricci, Francesco, Lior Rokach, and Bracha Shapira. 2015. Recommender Systems Handbook. Springer-Verlag. Vol. 54.<br>doi:10.1007/978-0-387-85820-3.                                                                           |
| book 2    | Bibliography | Brusilovsky, Peter, Alfred Kobsa, and Wolfgang Nejdl. 2007. The Adaptive Web: Methods and Strategies of Web Personalization. The Adaptive Web. Vol. 4321. doi:10.1007/978-3-540-72079-9.                                     |
| Article 2 | Bibliography | O'Donnell, E., Lawless, S., Sharp, M., Wade, V. (2015) A Review of Personalised E-Learning: Towards Supporting Learner Diversity. International Journal of Distance Education Technologies, 13(1), 22-47, January-March 2015 |
| book 3    | Bibliography | K. Falk. Practical Recommender Systems. 2019. Publisher: Manning Publications                                                                                                                                                |



## 9. Other information

---

### 9.1. Other information about the subject

To contact professors, students can use their email addresses included in this document or a Moodle message.

Professors will publish the teaching materials (slides, assignments, etc.) they use throughout the course on the Moodle site of the subject. Additionally, professors will use the Moodle forum to announce key events and provide relevant information on the subject.

Several innovative teaching methodologies are implemented in this course (<https://innovacioneducativa.upm.es/guias-pdi>) with the goal of motivating and reinforcing student learning:

- **Project-Based Learning:** This is a learning methodology in which students, working in small groups, are asked to plan, create, and evaluate a project that addresses the needs presented in a given situation.
- **Learning by Doing:** This is an approach that gives students an active role in the tasks to be carried out, granting them autonomy to resolve the given situation or problem.
- **Problem-Based Learning:** Students, working in teams, are required to complete two programming assignments based on real-world problems. They must apply the knowledge acquired in the course to solve these problems under the supervision of the instructors. Both assignments begin with two workshops in the classroom, respectively, in which the professor clarifies the goals of the assignments and provides support with technical issues.